[TGRS 2025] SeqCSIST: Sequential Closely-Spaced Infrared Small Target Unmixing

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arXiv: https://arxiv.org/abs/2507.09556

code: https://github.com/GrokCV/SeqCSIST

SeqCSIST: Sequential Closely-Spaced Infrared Small Target Unmixing

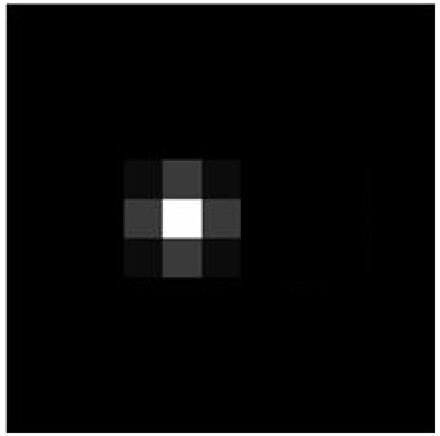
*1、**Why** Closely-Spaced Infrared Small Target Unmixing?*

*2、**How** to achieve Closely-Spaced Infrared Small Target Unmixing?*

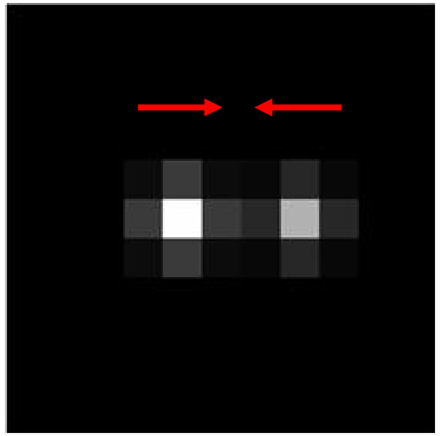
*3、**Specific** implementation*

*4、**Summary***

Why Closely-Spaced Infrared Small Target Unmixing?



One Target

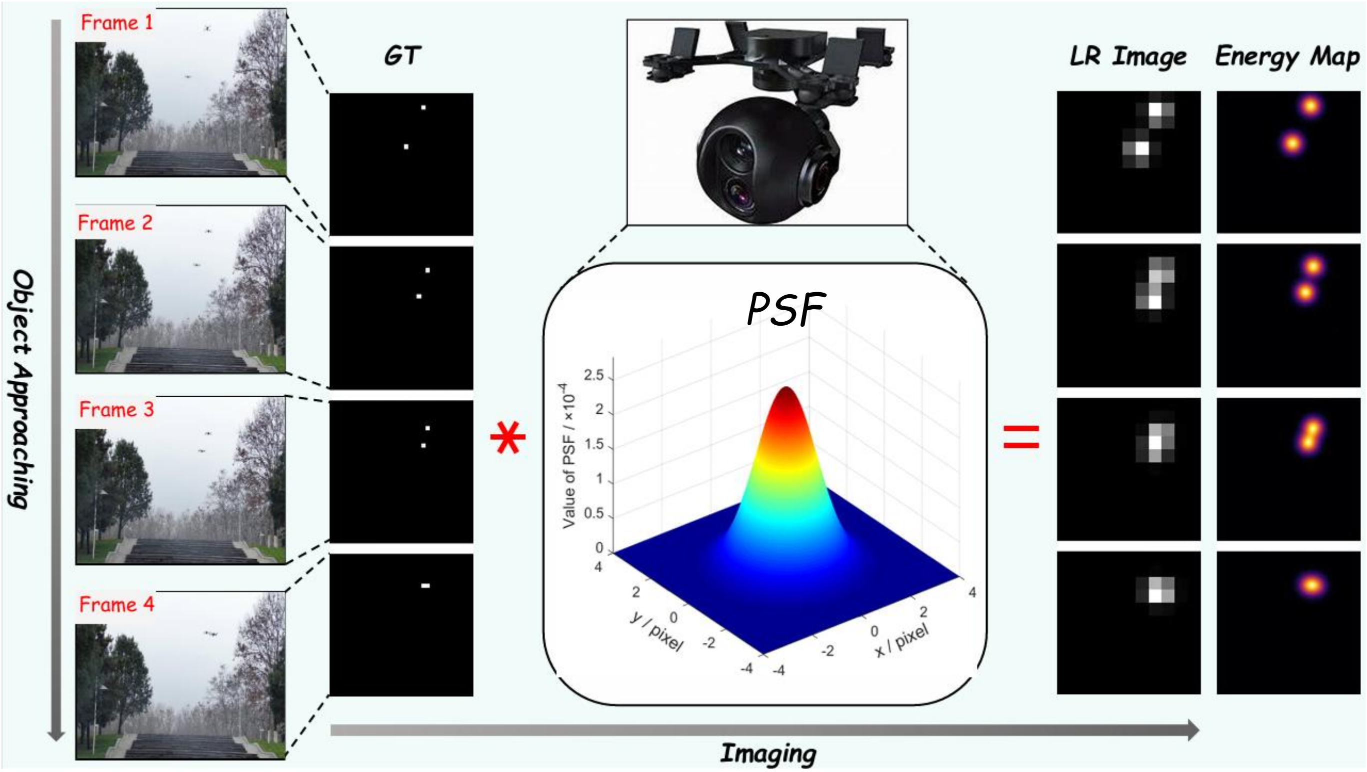


Two Targets

- ✓ *Energy Diffusion*
- ✓ *Energy Overlap*

What if the target distance is smaller?

Why Closely-Spaced Infrared Small Target Unmixing?

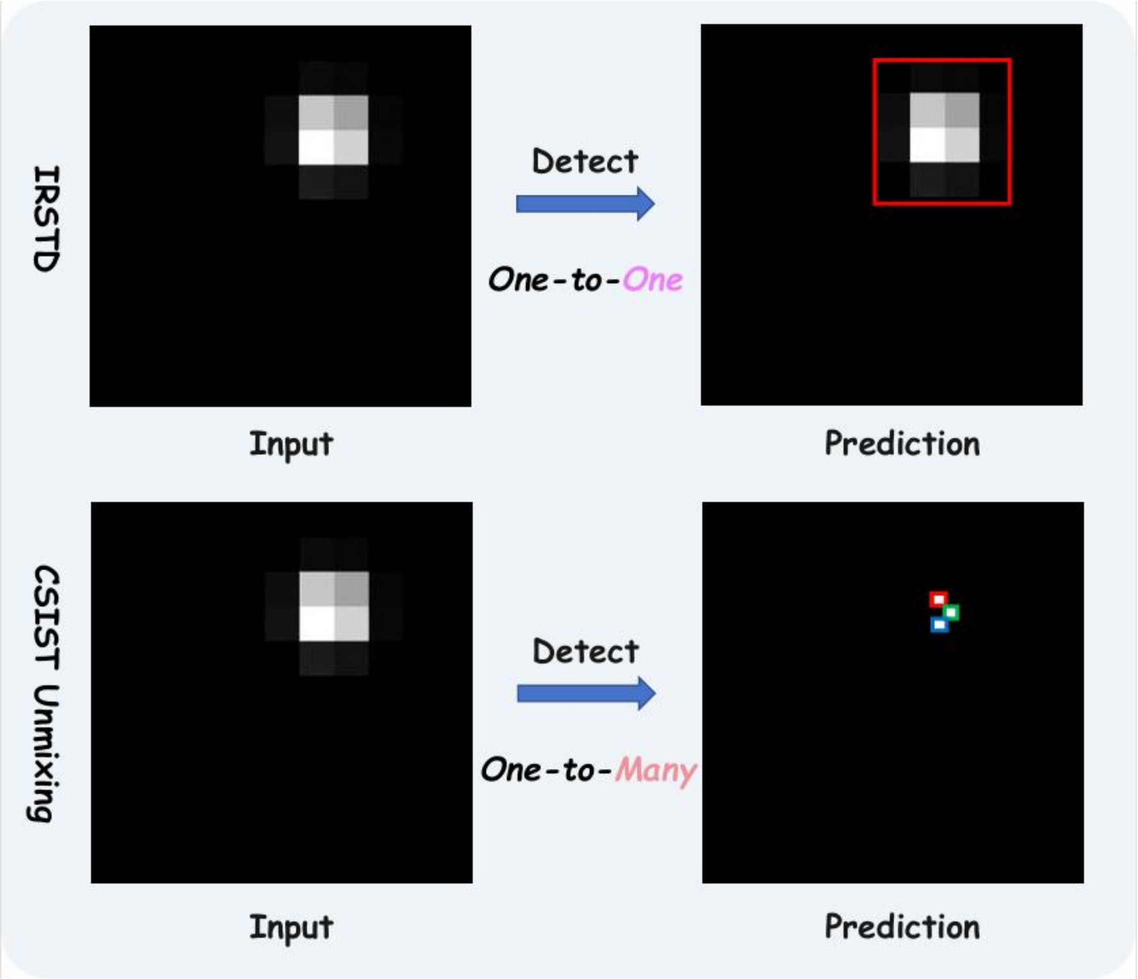


- ✓ *Energy Diffusion*
- ✓ *Energy Overlap*

The limitation of infrared detector's resolution

Difficult to distinguish and identify individual targets !

Why Closely-Spaced Infrared Small Target Unmixing?

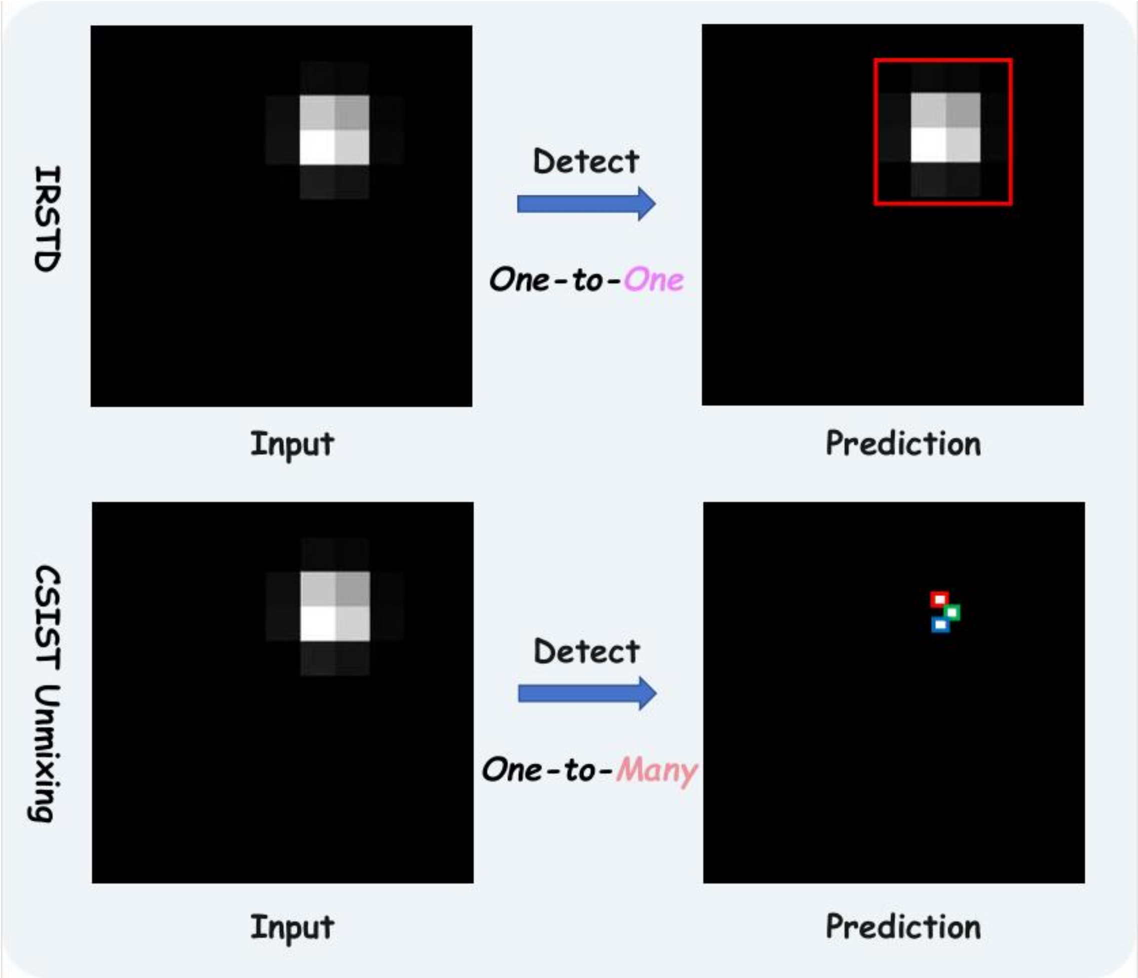


Infrared Small Target Detection

- ✓ *Limited to scenes of large field-of-view but sparse targets*
- ✓ *Assumes a one-to-one mapping between image objects and real targets*

Unable to distinguish and identify individual targets !

Why Closely-Spaced Infrared Small Target Unmixing?



★ **Novel Task**

Closely-Spaced Infrared Small Target Unmixing

- ✓ *Distinguishing closely packed, overlapped targets*
- ✓ *Transforming the task from simple detection to precise sub-pixel localization*

Focusing on resolving target groups that are visually indistinguishable!

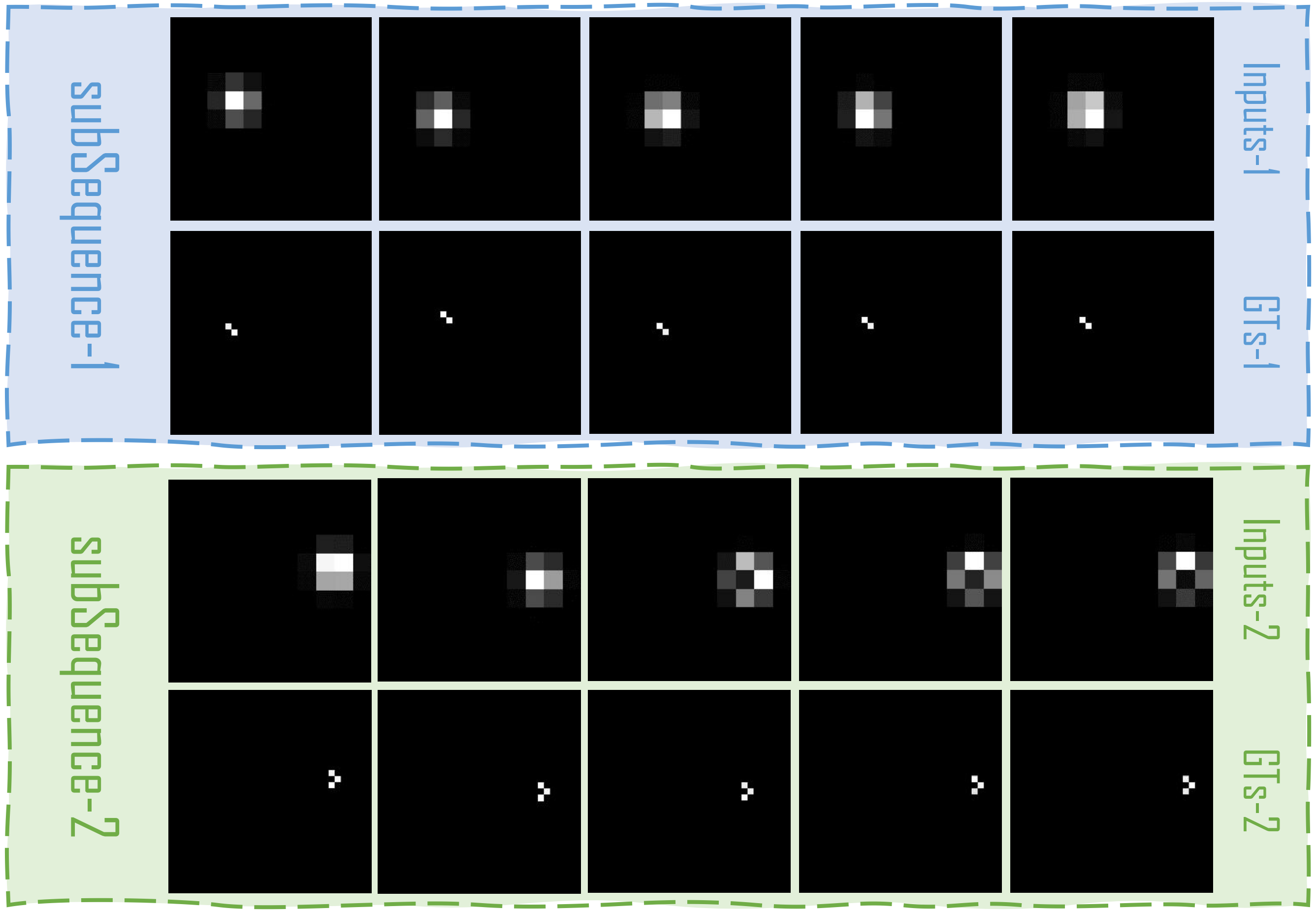
How to achieve?

IRSTD		CSIST Unmixing
Generic ResNet backbones	*Lacks task-specific priors* →	✓ *sparsity-driven feature extraction*
Complex spatial-temporal fusion module	*Complex and computation-intensive* →	✓ *Lightweight absolute position encoding*
Rely on optical flow method for inter-frame alignment	*weak spatial target awareness* →	✓ *Multi-frame deformable alignment*

Talent is limited without resources !

SeqCSIST Dataset

★ ***SeqCSIST** generation guided by diffraction and resolution theory*

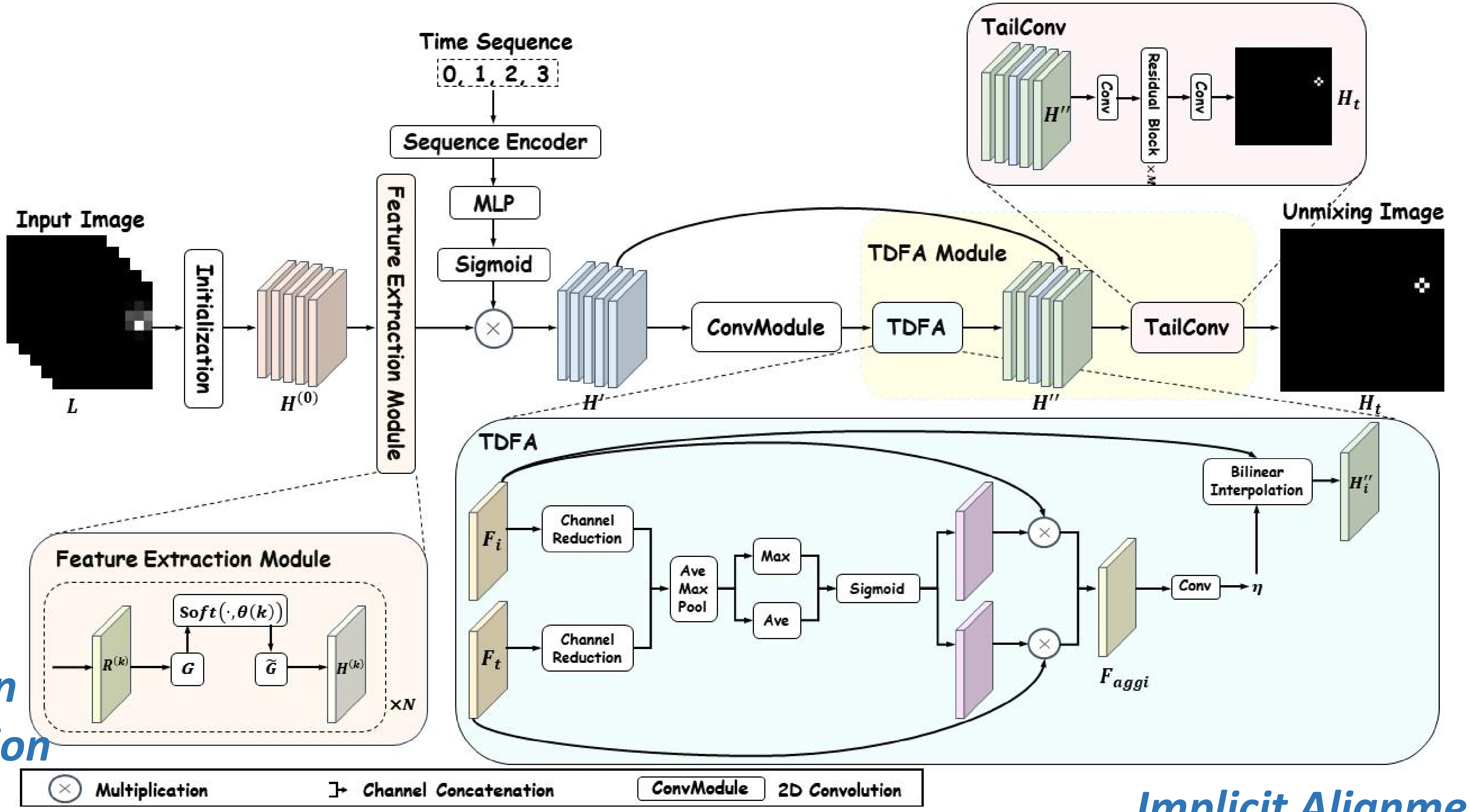


Category	*Value / Range*
Total frames	*100,000*
Number of trajectories	*5,000*
Frames per trajectory	*20 consecutive target crops*
Intensity Range	*220~250*
Overlapping target count	*2~4/pixel*
σ_{PSF}	*0.5 pixel*

Method

★ DeRefNet

Position Encoder



Sparsity-driven Feature Extraction

Implicit Alignment

Experiment

Comparison with State-of-the-Arts

Metric : *CSO-mAP*

★ ***A** **toolkit** **that provides objective evaluation metrics for this special task, along with the implementation of 23 relevant methods.***

Method	Params ↓	FLOPs ↓	CSO-mAP ↑					
			CSO-mAP	AP_{05}	AP_{10}	AP_{15}	AP_{20}	AP_{25}
Traditional Optimization								
ISTA [27]		398.57 M	10.72	0.14	1.97	8.74	18.22	24.53
BID [28]		10.89 M	14.40	0.00	3.00	13.00	26.00	30.00
Image Super-Resolution								
SRCNN [29]	15.84 K	0.35 G	49.64	1.40	16.30	51.20	85.00	94.30
GMFN [30]	2.80 M	27.53 G	50.94	0.70	11.90	51.20	92.10	98.80
DBPN [31]	1.96 M	4.75 G	50.40	0.80	12.50	51.20	90.00	97.40
SRGAN [32]	35.31 M	40.274 G	26.96	0.30	3.90	19.40	46.90	64.30
BSRGAN [33]	36.06 M	0.266 T	33.21	0.40	6.10	27.50	57.20	74.90
ESRGAN [34]	50.45 M	0.375 T	36.86	0.40	6.00	30.30	66.80	80.70
RDN [35]	22.31 M	53.97 G	49.61	0.70	10.60	48.20	90.40	98.20
EDSR [36]	0.39 M	0.99 G	50.19	0.60	10.30	48.80	92.20	99.00
ESPCN [37]	54.75 M	22.73 K	47.18	1.60	15.30	46.60	80.30	92.00
TDAN [26]	0.59 M	2.179 G	47.96	0.50	8.60	43.80	89.30	97.50
Deep Unfolding								
LIHT [38]	21.10 M	0.42 G	6.36	0.10	1.00	4.30	10.40	16.00
LAMP [39]	2.13 M	86.97 G	9.09	0.10	1.50	6.50	15.00	22.30
ISTA-Net [18]	0.17 M	4.09 G	48.95	0.70	11.20	49.70	87.70	95.40
FISTA-Net [40]	74.60 K	6.02 G	50.61	1.00	12.60	51.40	90.70	97.30
ISTA-Net+ [18]	0.38 M	7.70 G	51.02	1.00	13.70	52.70	90.40	93.70
ISTA-Net++ [19]	0.76 M	16.54 G	50.50	0.70	10.40	49.20	92.8	99.40
LISTA [41]	21.10 M	0.42 G	9.39	0.10	1.70	6.90	15.40	22.70
USRNet [16]	1.07 M	11.26 G	49.25	0.70	9.80	46.60	91.20	98.90
TiLISTA [42]	2.22 M	86.97 M	13.52	0.20	2.10	9.50	22.60	33.30
RPCANet [43]	0.68 M	14.81 G	47.17	0.70	10.20	44.50	84.60	95.90
⋆ **DeRefNet (Ours)**	0.89 M	15.70 G	<u>51.55</u>	1.00	14.40	54.90	90.40	97.10

Experiment

Ablation Study

TABLE II
PERFORMANCE OF FEATURE EXTRACTION MODULE AND STACKED RESIDUAL BLOCKS STRUCTURE

Backbone	CSO-mAP	AP_{05}	AP_{10}	AP_{15}	AP_{20}	AP_{25}
ResBlocks [26]	47.96	0.50	8.60	43.80	89.30	97.50
Deep Unfolding	**50.27**	**0.70**	**11.40**	**50.80**	**90.60**	**97.90**

TABLE III
SYSTEMATIC MODULE-WISE ABLATION STUDY:
IMPACT ASSESSMENT OF CORE COMPONENTS ON DEREFNET PERFORMANCE AND COMPUTATIONAL COST

Deep Unfolding	Optical Flow	DA	Time Encoder	DDA	Paras	FLOPs	CSO-mAP	AP_{05}	AP_{10}	AP_{15}	AP_{20}	AP_{25}
✓	✓	-	-	-	/	/	50.55	0.70	11.20	49.50	92.50	98.80
✓	-	✓	-	-	0.23 M	6.44 G	50.67	0.80	11.90	50.90	91.50	98.30
✓	-	✓	✓	-	/	/	51.39	0.80	12.40	52.20	92.70	98.80
✓	-	-	-	✓	0.28 M	6.14 G	51.09	0.80	12.00	52.00	92.30	98.40

Experiment

Generalization experiment

TABLE IV
DEREFNET PERFORMANCE ON SEQCSIST AND HYBRID DATASETS WITH HYBRID REFERRING TO SIMULATED CSOS COMPOSITED ONTO REAL-WORLD INFRARED SCENES

DATASET	CSO-mAP	AP_{05}	AP_{10}	AP_{15}	AP_{20}	AP_{25}
SeqCSIST	**51.55**	**1.00**	**14.40**	**54.90**	**90.40**	**97.10**
Hybrid Dataset	47.48	0.70	11.00	46.30	84.80	94.60

TABLE V
COMPREHENSIVE NOISE TOLERANCE ASSESSMENT: DEREFNET PERFORMANCE ANALYSIS UNDER ADDITIVE GAUSSIAN NOISE WITH DIFFERENT STANDARD DEVIATION VALUES

σ	CSO-mAP	AP_{05}	AP_{10}	AP_{15}	AP_{20}	AP_{25}
0 (clean)	**51.55**	**1.00**	**14.40**	**54.90**	**90.40**	**97.10**
2	49.09	0.80	12.40	50.50	86.60	95.30
3	48.40	0.80	11.80	48.00	86.00	95.30
4	48.41	0.90	12.20	48.00	85.80	95.20
5	47.23	0.80	11.40	46.10	84.10	93.90

Visualization

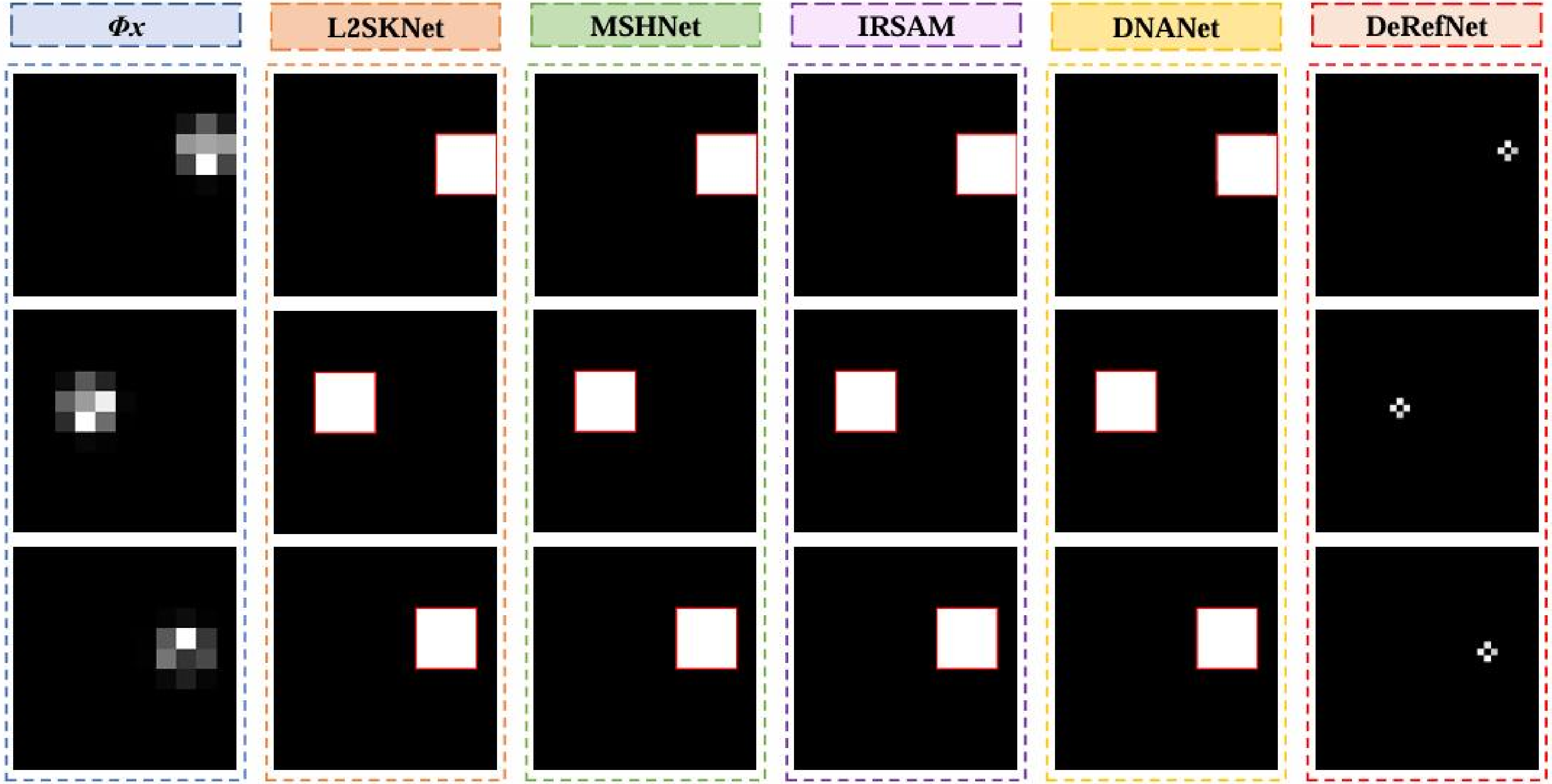
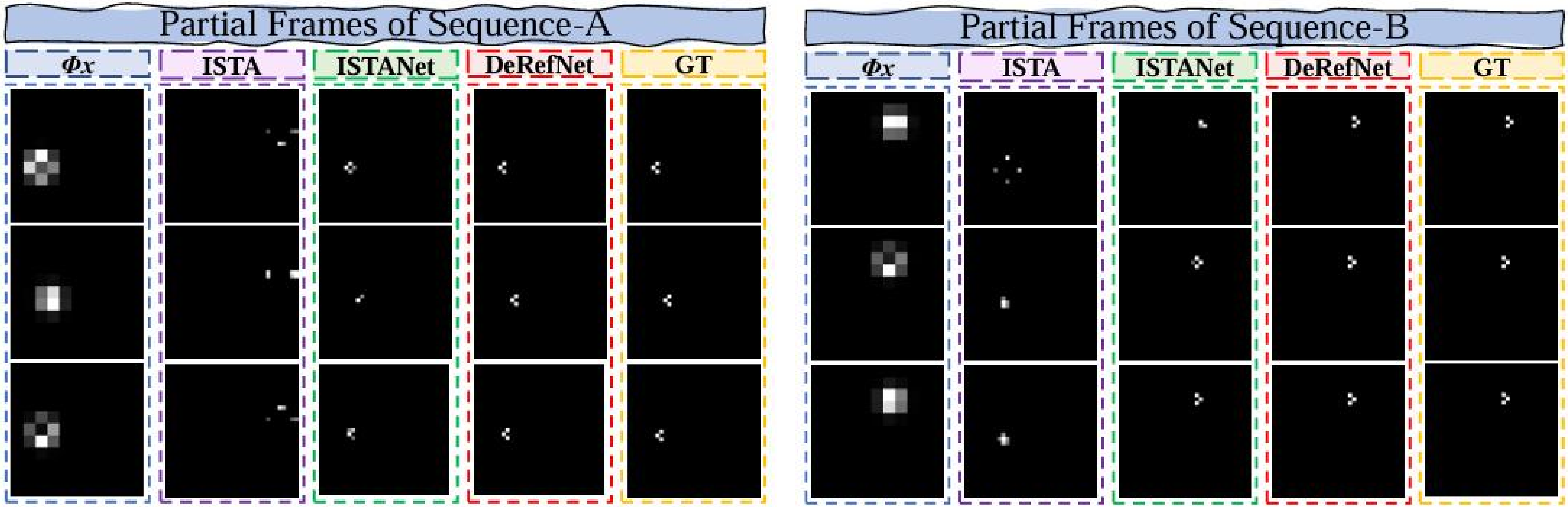


Fig. 5. Partial frame unmixing effect of subsequences of different trajectories. The visualization demonstrates DeRefNet's superior capability in resolving closely-spaced infrared small targets, where individual target signatures and precise sub-pixel positions are successfully extracted from densely clustered regions that conventional detection methods can only provide as coarse binary mask results.

Visualization



Summary

This work is the first endeavor to address the CSIST Unmixing task within a multi-frame paradigm.

Contribution

***Novel Task**: We propose a novel task, Sequential CSIST Unmixing, which broadens the definition of IRSTD*

***Open-Source Ecosystem:** We release an open-source ecosystem for CSIST Unmixing, including the SeqCSIST dataset and a toolkit for benchmarking*

***End-to-End Framework:** We propose DeRefNet, a multi-frame model-driven deep learning architecture with temporal deformable feature alignment.*

THANK YOU!

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code: https://github.com/GrokCV/SeqCSIST